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SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

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Note

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IMAGE-TO-TEXT

Note

[illegible]

-  Simone Scannapieco

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- ## Utilizzo GPT-4.0 per visione computazionale

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Note

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- 1 Creazione interfaccia ed implementazione del servizio
- 2 Modifica controllore MVC
- 3 Test delle funzionalità con Postman/Insomnia

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```
package it.venis.ai.spring.demo.services;

import org.springframework.core.io.Resource;

import it.venis.ai.spring.demo.model.Answer;

public interface MultiModalityService {

    Answer getTranscriptionFromAudioFile(Resource audioFile);

    Answer getDescriptionFromImage(Resource imageFile);

}
```

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface. There is no handwriting or other markings on the paper.

```
package it.venis.ai.spring.demo.services;

import javax.print.attribute.standard.Media;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;
    }
    ...
    @Override
    public Answer getDescriptionFromImage(Resource imageFile) {

        String contentType = getContentType(imageFile);

        Media imageMedia = new Media(MimeTypeUtils.parseMimeType(contentType), imageFile);

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Descrivi in lingua italiana la seguente immagine: ").media(imageMedia))
            .call()
            .content());
    }
}
```

Note

[illegible]


```
...
private String getContentType(Resource multimediaFile) {
    String contentType = URLConnection.guessContentTypeFromName(multimediaFile.getFilename());

    if (contentType == null || contentType.equals("application/octet-stream")) {
        String fileName = multimediaFile.getFilename();
        if (fileName != null) {
            String extension = fileName.substring(fileName.lastIndexOf('.') + 1).toLowerCase();
            contentType = switch (extension) {
                case "mp3" -> "audio/mpeg";
                case "wav" -> "audio/wav";
                case "m4a" -> "audio/mp4";
                case "ogg" -> "audio/ogg";
                case "flac" -> "audio/flac";
                case "jpg" -> "image/jpeg";
                case "jpeg" -> "image/jpeg";
                case "png" -> "image/png";
                case "raw" -> "image/raw";
                default -> "media/unknown";
            };
        } else {
            contentType = "media/unknown";
        }
    }

    return contentType;
}
}
```

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```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@RequestMapping("/gemini/multi-modality")
public class GeminiMultimodalityController {

    private final MultiModalityService multiModalityService;

    public GeminiMultimodalityController(MultiModalityService multiModalityService) {

        this.multiModalityService = multiModalityService;
    }

    ...

    /**
     * Endpoint per l'analisi e descrizione di immagini (Image-to-Text).
     * Carica un'immagine predefinita dal classpath e utilizza le capacità di visione
     * di Gemini per generare una descrizione testuale del contenuto dell'immagine.
     *
     * @param audioFile risorsa immagine dal classpath (file PNG predefinito)
     * @return Answer oggetto contenente la descrizione generata dell'immagine
     */
    @PostMapping("/itt")
    public Answer getDescriptionFromImage(@Value("classpath:multimodal.test.png") Resource audioFile) {

        return this.multiModalityService.getDescriptionFromImage(audioFile);
    }

}
```

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